ConceptOS - Product Requirements Document (PRD)

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Project: Adaptive Exam Intelligence System

Codename: ConceptOS

1. Product Vision

Build an AI-powered adaptive learning platform that teaches students through previous year question patterns instead of traditional chapter-first learning.

The system analyzes:

• examiner intent

• historical question trends

• concept recurrence

• student cognitive behavior

to generate personalized preparation journeys for competitive exams like JEE and NEET.

2. Problem Statement

Current edtech platforms suffer from:

• content overload

• generic learning paths

• weak personalization

• lack of concept prioritization

• poor understanding of examiner behavior

Students waste time studying:

• low-yield concepts

• irrelevant theory

• non-recurring patterns

3. Core Product Thesis

Traditional Learning:

Theory → Notes → Questions

ConceptOS Learning:

Questions → Concept Extraction → Pattern Recognition → Adaptive Reinforcement

4. Target Users

Primary users:

• JEE aspirants

• NEET aspirants

• Future UPSC/GATE aspirants

5. Core Features

A. PYQ Intelligence Engine

Analyzes previous year questions to identify:

• recurring concepts

• difficulty

• weightage

• examiner intent

B. Concept Graph Engine

Builds interconnected concept relationships and dependency mapping.

C. Student Cognitive Model

Tracks:

• solving speed

• conceptual errors

• retention decay

• confidence

• learning style

D. Adaptive Teaching Engine

Dynamically chooses:

• next question

• explanation style

• revision schedule

• reinforcement timing

E. Examiner Brain Mode

Explains:

• hidden examiner patterns

• trick logic

• concept traps

F. AI Revision Planner

Generates adaptive daily schedules based on:

• exam date

• target percentile

• weak concepts

• available study hours

G. Self-Improving Agent Layer

Uses reinforcement learning from:

• score improvements

• retention gains

• engagement patterns

6. System Architecture

Frontend:

• Next.js

• React

• Tailwind CSS

Backend:

• Python FastAPI

• Node.js

AI Stack:

• PyTorch

• LangChain

• LlamaIndex

Databases:

• PostgreSQL

• Pinecone/Weaviate

• Neo4j

Infrastructure:

• AWS/GCP

• Redis

• Kafka

• Kubernetes

7. MVP Scope

Initial focus:

• JEE Physics only

MVP includes:

• PYQ ingestion

• concept tagging

• adaptive sequencing

• AI explanations

• student profiling

• revision planner

8. Monetization

• Freemium model

• Premium adaptive mentor

• Institutional licensing

• Coaching center partnerships

9. Competitive Advantage

ConceptOS differs through:

• Question-first learning

• Cognitive adaptation

• Examiner intelligence

• Self-improving AI learning loops

10. Future Vision

ConceptOS evolves into:

- a fully autonomous AI mentor
- a personalized learning operating system
- a continuously self-improving educational intelligence platform